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PI Name:	Lelkes, Peter Istvan Ph.D.		
Project Title:	Altered Signal Transduction and Differential Gene Expression in PC12 Pheochromocytoma Cells Cultured in RWV Bioreactors		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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PI Email:	pilelkes@temple.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-204-3307
Organization Name:	Temple University		
PI Address 1:	Department of Bioengineering		
PI Address 2:	1947 N 12th St, Room 811		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19122-6018	**Congressional District:**	2
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	The focus of this ground-based proposal is to characterize signaling mechanisms and differential gene expression in PC12 pheochromocytoma cells cultured in NASA's Rotating Wall Vessel (RWV) Bioreactors, a low-shear, microgravity (microG)- simulating cell culture venue. PC12 cells, neural crest derived and bipotential, are well established cellular models for studying signal transduction involved in neuronal/neuroendocrine differentiation Preliminary data suggest alterations in cellular signaling patterns and differential gene expression in PC12 cell maintained in RWV Bioreactors for only short periods of time (hours). Based on these observations it is hypothesized that the microgravity-simulating cell culture environment in NASA's RWV Bioreactors is sufficient to induced differential gene expression through alterations in phosphorylation-dependent signaling pathways. The long-term goal of this proposal is to provide NASA with a well-characterized system, responsive to modeled microgravity, which will be ready for further testing on board the International Space Station. The objective of this proposal is to 1. Study the time		

Task Description:	course of the acquisition of a neuroendocrine/adrenergic phenotype in PC12 pheochromocytoma cells cultured in simulated microgravity. 2: Study the time course of the down-regulation of the neuronal phenotype in PC12 cells cultured in simulated microgravity. 3: To investigate the cellular and molecular mechanisms by which simulated microgravity alters protein phosphorylation dependent signaling pathways. 4: Establish a causal link between simulated microgravity-induced alteration in cellular signaling and PNMT activation. The long-term goal of this proposal is to provide NASA with a well-characterized system, responsive to modeled microgravity , which will be ready for further testing on board the International Space Station. This study has contributed to the characterization and elaboration of NASA-designed RWV Bioreactors as a unique, microgravity-simulating cell culture environment here on earth. Also, this unique "in vitro embryology" venue has helped to identify unique signaling pathways involved in the crucial steps necessary for the formation of differentiated tissue-like assemblies from single cells and cell aggregates. The technological advances of the real time optical monitoring system will be useful for designing a rational, fully automated feedback mechanism for controlling the culture conditions (e.g., rotational speed) in RWV Bioreactors. These Bioreactors, developed by NASA and produced under NASA license have become one of the most exciting and innovative cell culture venues world-wide.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will contribute to the characterization and elaboration of NASA-designed RWV Bioreactors as a unique, microgravity-simulating cell culture environment here on earth. Also, this unique "in vitro embryology" venue will help to identify unique signaling pathways involved in the crucial steps necessary for the formation of differentiated tissue-like assemblies from single cells and cell aggreates.
Task Progress:	Significant progress has been achieved in all specific aims. We have further refined our understanding of the time course for the acquisition of a neuroendocrine/adrenergic phenotype in PC12 pheochromocytoma cells cultured in simulated microgravity. Specifically, we solidified and extended previous observations that within the first two hours upon culture in the RWV environment, certain signaling pathways, necessary and sufficient to induce the neuroendocrine / adrenergic phenotype, are upregulated. We have extended these observations now to experiments lasting up to 20 days. We found that the neuroendcirne phenotype, as assessed by PNMT expression is fully developed within the first 72 hours of RWV culture and maintained over the full culture period. Confirming our initial observations we find that many of the signaling pathways that are unregulated early on (within the first 2-24 h) remain activated for the entire duration of the RW culture Examples for these pathways are second messenger systems (cAMP), signaling through three of the mitogen activated kinases and specific nuclear transcription factors, such as egr-1, GRE and CREB. Interestingly, the expression of the low affinity neurotrophin receptor p75, but not of the high-affinity receptor trk, is also upregulated in PC12 cells cultured in RWV Bioreactors. Again, this is an effect that is observed on both the gene and the protein level within the first 2-4 hours and continues through the entire experimental duration. Normally, under 1 g condition, exposure to neurotrophins such as NGF leads to the rapid transient activation of cellular signaling via trk phosphorylation and subsequent activation of mitogen activated receptor pathway, resulting in the expression of a neuronal phenotype. In the modeled microgravity environment of the microgravity RWV Bioreactors, NGF-induced trk phosphorylation is inhibited while some of the down stream signaling molecules are switched on gradually (over a 2-6 h period), but then remain “in continual high gear”, independently of the presence or absence of NGF. Examples for this class of signaling molecules include members of the mitogen activated kinase families, notably mek-1 and erk 1 and erk 2, Similarly, and in contrast to 1g conditions, other member of the “side branches” of the family of mitogen activated kinases, p38 and jnk, are also “spontaneously” activated over extended time periods in RWV cultures. The altered pattern of signaling in RWV Bioreactors has now also been confirmed using gene expression profiling as well as global protein expression analysis using 2-D gel electrophoresis. Given the large scope of these expression profiles, we continue to mine our data bases in order to obtain meaningful correlations and to be able to establish clustering of signaling molecules. For the quantitative analysis of these huge data sets, and for establishing functional correlations, we are using state-of-the-art bioinformatics approaches and databases (such as the open-access data bases DAVID, KEGG, GO and, most recently, the commercial “gold-standard” Ingenuity). This level of sophisticated data mining will help us to elucidate how alterations in the signaling pathways, as identified through our experiments may related to the altered biological responses seen in the modeled microgravity environment in the RWV bioreactors, and ultimately during space flight. In order to elucidate some of the cellular and molecular mechanisms by which the modeled microgravity environment in the RWV Bioreactors alters protein phosphorylation-dependent signaling pathways, we used established inhibitors and activators which abrogate some of these pathways. In extension of our previous work, we established that the combined inhibition of signaling pathways down stream from the p75 low-affinity neurotrophin receptor in conjunction with inhibition of signals downstream of p38 is necessary and sufficient to prevent the induction of PNMT (gene/protein /activity). Importantly, our data suggest that alterations in several, seemingly independent signaling pathways are required to abrogate the differentiation towards the neuroendocrine phenotype of PC12 cells cultured in simulated microgravity in RWV Bioreactors. In addition to studying the activation/phosphorylation of individual proteins, we are also utilizing 2-D gel electrophoresis technology as a novel approach towards global assessment of protein phosphorylation - dependent signal transduction mechanisms. With this novel approach we are capitalizing on or past experience in analyzing phosphorylation -dependent signaling by 1-D gel electrophoresis. Using a phosphoprotein-specific stain, we have been able to identify dozens of proteins, which are differentially phosphorylated in the various venues. The focus of current and future efforts is to analyze differential phosphoprotein activation/ expression , as a commensurate of cellular signaling. As before, we are using complex bioinformatics tools to elucidate differential activation of signaling pathways in the various venues. Knowledge of global signaling events common between the various venues and/or unique to the RWV environment, will allow us to design effective therapeutic targets to treat unique physiological disturbances, which will be useful for designee effective countermeasures at the molecular level. In extending prior observation, by us and others, we determined that culture in RWV leads to an activation of cAMP – dependent signaling. Indeed, specific inhibition of protein kinase A leads to an abrogation of PNMT induction in

	simulated microgravity. Conversely, using specific activators of protein kinase A and hence, of cAMP dependent signaling pathways, we have demonstrated for the first time that activation of this pathways leads to and accelerated and end enhance neuroendocrine differentiation of PC12 cells, as assessed by the an significant increase in the expression of PNMT. Finally, as a new technological advance, we developed a novel optical system for determining in real-time the aggregation kinetics of cells cultured in RWV Bioreactors. The long-term goal of this engineering effort is to use the optical data for obtaining detailed biological information from the parameters that affect the kinetics of aggregation. Technologically, the data will also serve as the basis for developing feed-back algorithms, which will allow us to generate algorithms for automatically controlling the rotational speed and other vital parameters of RWV Bioreactors, based on the real-time monitoring of the aggregation process. We anticipate that the novel methodology and algorithms developed in this work will enhance our ability for non-invasive, remote monitoring and data analysis during the exploration to Moon and Mars. The results form these studies form the basis for several publications which are currently being prepared for publication in peer-reviewed journal. Taken together, in line with the original goal and specific aims of this grant, our data have helped to establish a causal link between modeled microgravity in RWV -induced alteration in cellular signaling and PNMT activation and provided experimental basis towards our long-term goal, namely to provide NASA with a well-characterized cellular system, responsive to simulated microgravity, which might be ready for further testing on board the International Space Station or during the subsequent stages of space exploration .
Bibliography Type:	Description: (Last Updated: 09/28/2023)
Articles in Peer-reviewed Journals	Dollé JP, Rezvan A, Allen FD, Lazarovici P, Lelkes PI. "Nerve growth factor-induced migration of endothelial cells." J Pharmacol Exp Ther. 2005 Dec;315(3):1220-7. Epub 2005 Aug 25. PMID: 16123305 , Dec-2005
Dissertations and Theses	Simons DM. "Recovery of ConA-Induced Activation by Modeled Microravity-exposed Balb/c Mouse Splenocytes." Drexel University, December 2004. , Dec-2004
Dissertations and Theses	Frank, Jennifer D. "Global Analysis of P12 Pheochromocytoma Cell Phosphoproteins under Acute NGF Stimulation by 2-D SDS PAGE." Drexel University, May 2005. , May-2005
Dissertations and Theses	Manley P. "A Novel Real Time System to Monitor Cell Aggregation and Trajectories in RWV Bioreactors." Drexel University, August 2005. , Aug-2005